

CNS

By Dr.Riffat Ibrahim Riffat

02 Psychosis



Dopaminergic pathways in brain

Mesolimbic and mesocortical pathway:

- They are responsible for Cognition, memory, attention, behavior, and learning.
- Dopamine stimulates these functions.

Basal ganglia:

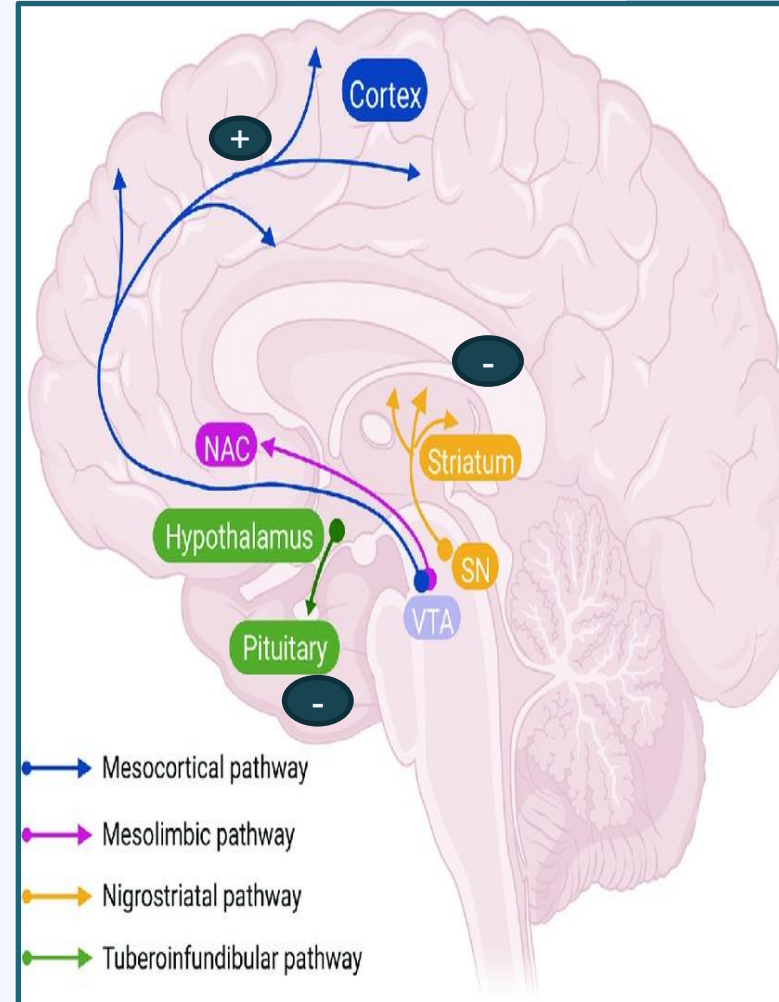
- Acts as inhibitory N.T to balance the stimulatory acetylcholine (regulation of fine movements)

On pituitary gland

- Suppresses prolactin secretion

On V.C:

- induces vomiting.



- **Psychosis or schizophrenia** is a mental illness that affects 1% of the population. psychosis has a strong genetic predisposition and is characterized by:

- **Positive symptoms:**

- Dopamine increase in mesolimbic and mesocortical track is responsible for these abnormalities.
- Series of delusions, as a set of ideas that the patient compulsively believes in and can not be convinced of their fallacy.
- Auditory and visual hallucinations.



- **Negative symptoms:**

- Serotonin is responsible for these manifestations.
- Depression, apathy, blunt emotions.
- Social isolation often resulting in lack of self-care And hygiene.



Overview

- The use of antipsychotic medications involves a difficult trade-off between the benefit of alleviating psychotic symptoms and the risk of a wide variety of troubling adverse effects.
- Antipsychotic drugs are not curative and do not eliminate chronic thought disorders, but they often decrease the intensity of hallucinations and delusions and permit the person with schizophrenia to function in a supportive environment.



First generation (Typical or Classic drugs)

- Phenothiazines class :Chlorpromazine (CPZ)
 - Butyrophenones class : Haloperidol and Droperidol .
- 

- **MOA:**

- Competitive inhibition of D2 receptors, however, they are not selective to mesolimbic and mesocortical D2.
- 





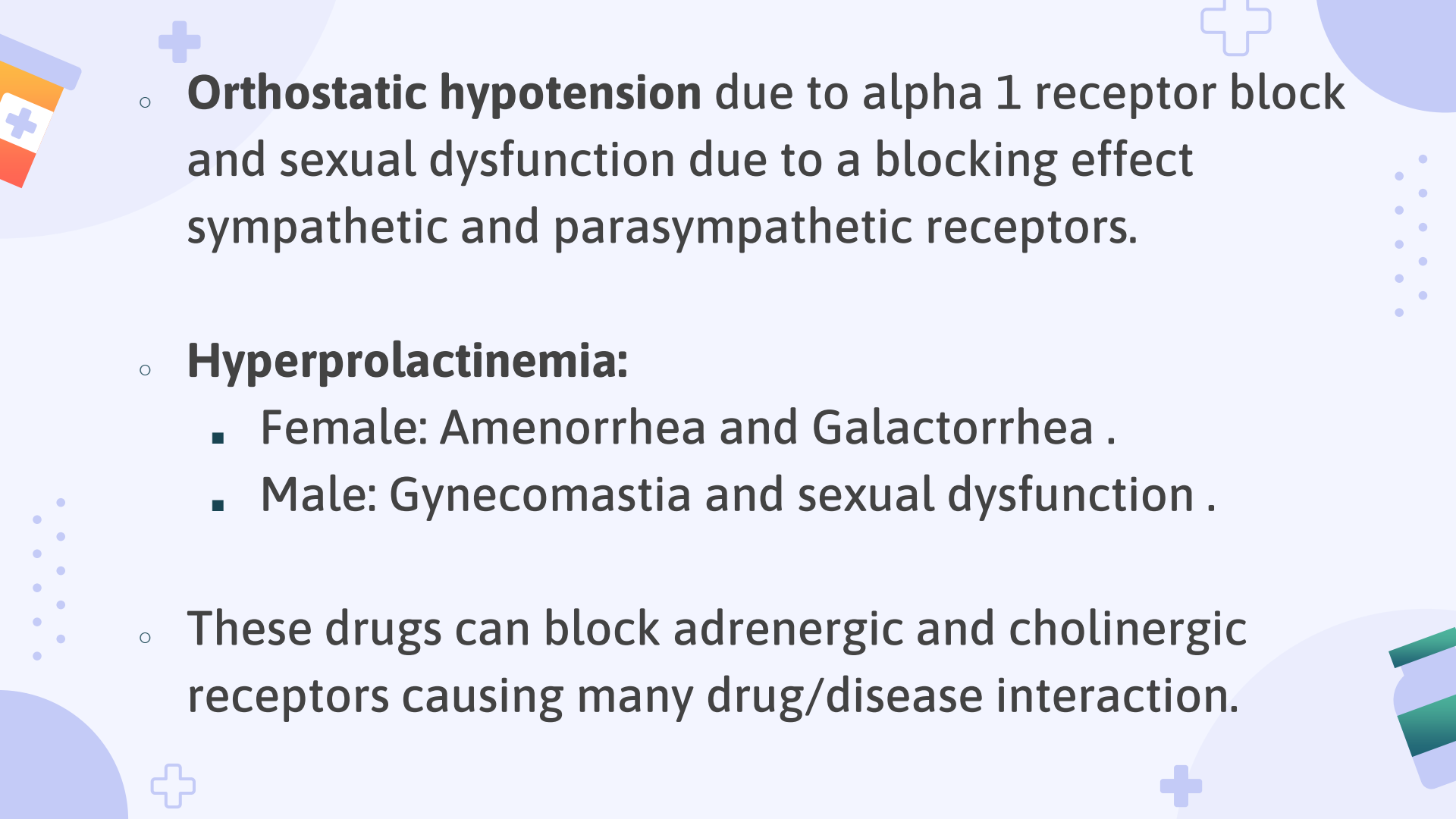
• Therapeutic uses:

- Treatment of psychosis and Schizophrenia (+ve symptoms).
 - In severe vomiting.
 - Chlorpromazine is effective for hiccup.
 - As tranquilizers in manic states.
- 

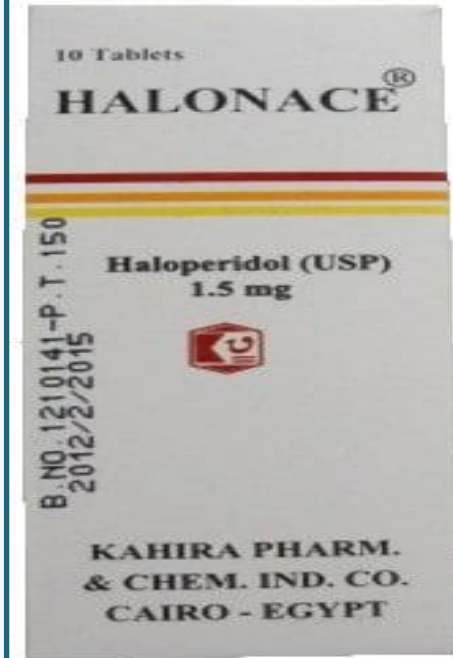
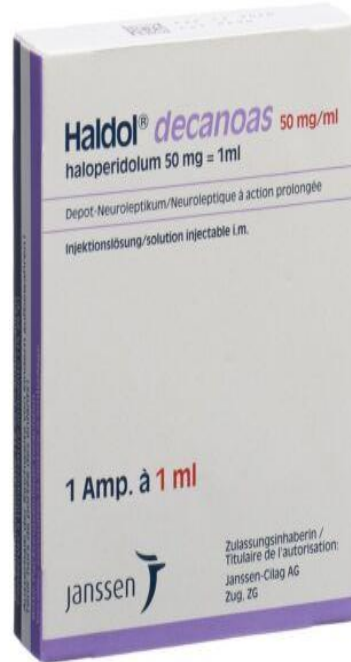


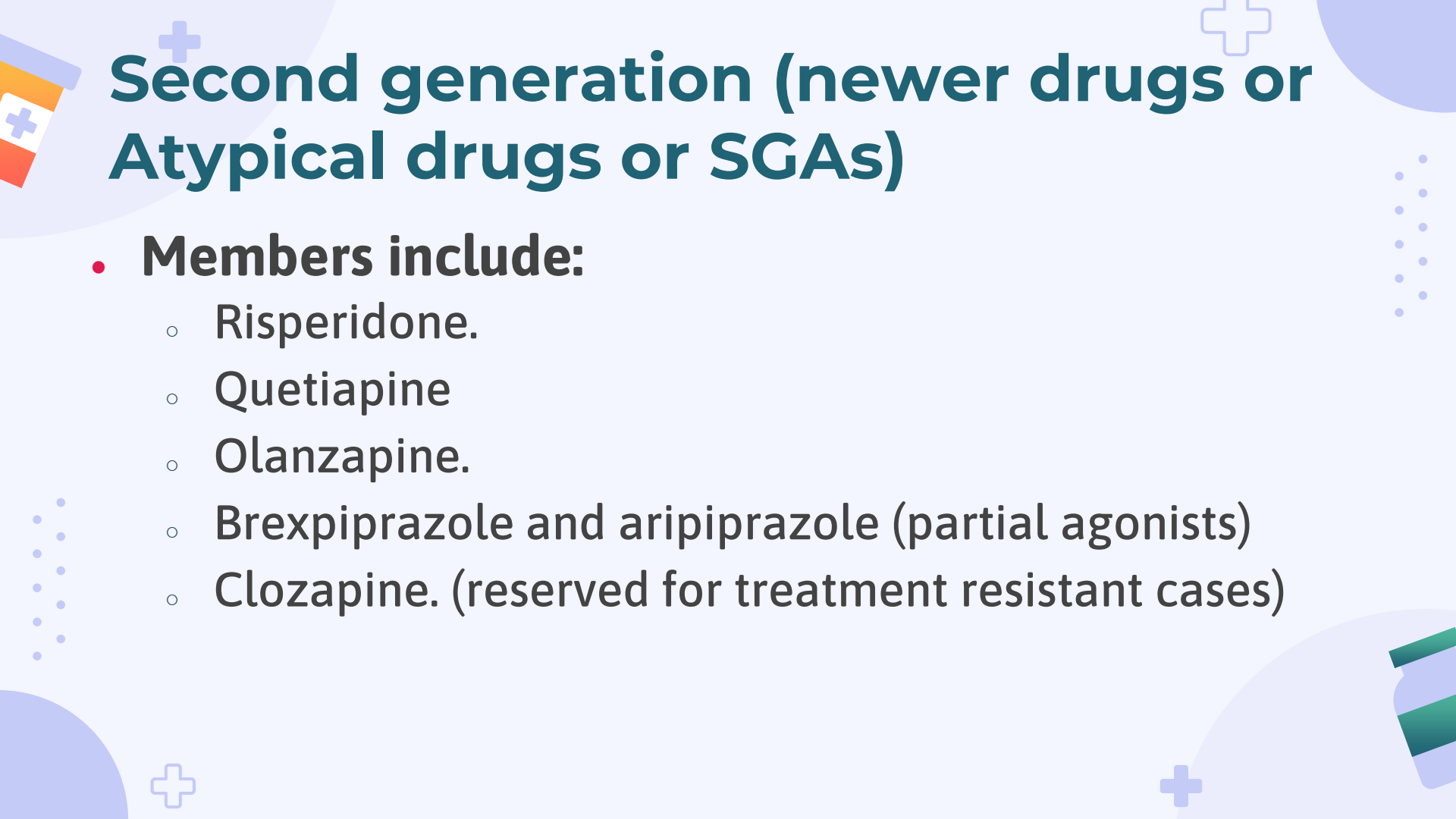
Adverse effects

- **Extrapyramidal side effects (EPS)** is a collection of CNS related side effects due to D2 receptor block in basal ganglia.
 - Akathisia which is a feeling of restlessness and anxiety (reversible)
 - Dyskinesia defined by abnormal body movements specially facial.
 - Dystonia abnormal uncontrolled body motions (irreversible)
- **Neuroleptic Malignant Syndrome (NMS):**
 - They are a set of autonomic dysfunctions characterized by:
 - Hyperthermia ,Convulsion ,Muscle rigidity ,hypertension and sweating.

- 
- **Orthostatic hypotension** due to alpha 1 receptor block and sexual dysfunction due to a blocking effect sympathetic and parasympathetic receptors.
 - **Hyperprolactinemia:**
 - Female: Amenorrhea and Galactorrhea .
 - Male: Gynecomastia and sexual dysfunction .
 - These drugs can block adrenergic and cholinergic receptors causing many drug/disease interaction.

Brands





Second generation (newer drugs or Atypical drugs or SGAs)

- **Members include:**

- Risperidone.
- Quetiapine
- Olanzapine.
- Brexpiprazole and aripiprazole (partial agonists)
- Clozapine. (reserved for treatment resistant cases)



• MOA:

- These drugs are antagonist of D2 receptors and 5-HT2A receptors.
 - Not all members show the same affinity to D2 and 5-HT2A receptor. Some have an equal affinity to both receptors while others prefer one target over the other.
 - The distributed affinity explains two effects of SGAs:
 - Reduced incidences of E.P.S since the drug acts on 2 targets rather than D2 receptors only.
 - SGAs improve treatment of –ve and +ve symptoms when compared to FGAs that treat only –ve symptoms. 
- 



Core Pharmacological Difference:

- Full Antagonists: These drugs (e.g., Haloperidol, Risperidone) block the D2 receptor. They sit on the receptor without activating it, preventing dopamine from binding. This directly reduces dopamine signaling in the mesolimbic pathway, which is thought to be overactive in psychosis.
- Partial Agonists: These drugs (e.g., Aripiprazole, Brexpiprazole, Cariprazine) act as a "dimmer switch." They have lower intrinsic activity than dopamine itself. In areas of high dopamine (like the mesolimbic pathway in psychosis), they act as functional antagonists by displacing the more potent dopamine and producing a weaker signal. In areas of low dopamine (like the mesocortical pathway, linked to negative/cognitive symptoms), they can provide a gentle boost to dopamine signaling.





• **Therapeutic uses:**

- Treatment of psychosis and schizophrenia (1st line).
 - As tranquilizers in manic states.
 - Risperidone is used in short-term treatment of persistent aggression in patients with moderate to severe dementia of the Alzheimer's
- 
- 
- 
- 



• Adverse effects:

- Weight gain, diabetes, and hypercholesterolemia due to 5-HT_{2A} block. This effect is milder with partial agonists but this also makes them weaker in severe cases or poor responders
- NMS
- Leukopenia: often fatal and is common with Clozapine.
- EPS can still occur with the atypical drugs although the risk is less than that of the classic drugs.
- Both first- and second-generation drugs can prolong QT interval making them torsadogenic drugs.



Brands





Maintenance treatment



- Patients who have had two or more psychotic episodes secondary to schizophrenia should receive maintenance therapy for at least 5 years, and some experts prefer indefinite therapy.
- Low doses of antipsychotic drugs are not as effective as higher-dose maintenance therapy in preventing relapse.
- The rate of relapse may be lower with second-generation drugs





+ Challenges with antipsychotic therapy

- Adherence to antipsychotic treatment can be challenging to patients for many reasons including:
 - Poor insight into the illness
 - Denial of the illness
 - Cognitive impairments
 - Adverse effect profile (80% clinically significant adverse effects)
 - The use of long-acting formulation (haloperidol retard inj) will not prevent all forms of non-adherence, however, it can give the clinician more insight in the patient's state.
 - When using FGAs or multiple SGAs a central acting anti-muscarinic agent maybe added to reduce tremors and Parkinson-like symptoms as long as it doesn't progress into dyskinesia. +
- 

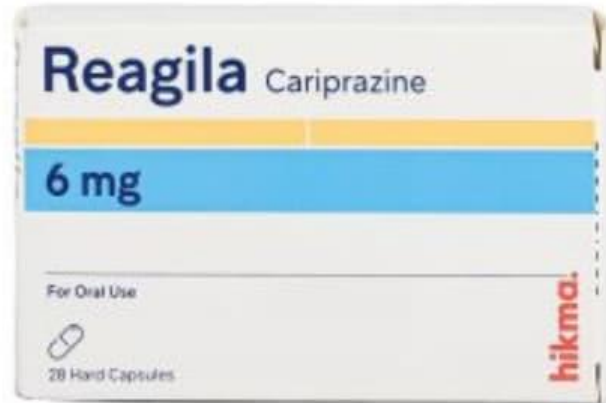
Further developments

- **Cariprazine** is a dopamine D2/D3 receptor partial agonist, with a distinctively high affinity for D3.
- This preferential D3 activity is unique among its class and may contribute to improvements in negative symptoms and cognitive function Psychiatry.
- It also modestly acts as a 5-HT1A partial agonist, and antagonizes 5-HT2A/2B and H1 receptors, while having minimal binding to 5-HT2C and α -adrenergic receptors, and negligible affinity for muscarinic cholinergic receptors
- Cariprazine is considered a third-generation anti-psychotic.

Further developments

- **Cariprazine is FDA-approved for:**
 - Schizophrenia (maintenance and symptom reduction)
 - Bipolar I disorder (manic or mixed episodes, and bipolar depression)
 - Major depressive disorder (MDD) as adjunctive therapy in unipolar depression (approved December 2022)
- Olanzapine and others are widely used in schizophrenia and bipolar mania, but not typically for depression.
- It has a low liability for metabolic effects—minimal weight gain, changes in glucose, lipids, prolactin, or QTc prolongation—all notably lower than with many SGAs like olanzapine or risperidone

Brands



 UnitedPharmacy0

 United.Pharmacy

Cases

- Mr S, a 25-year-old white pharmacy student, comes to your pharmacy in quite a distressed state. He is totally unkempt and smells of body odors; his mood is volatile and he reports to you that he has sensations of insects crawling under his skin. Over the last few days he has been feeling sick, and he is absolutely convinced that this is because his neighbor somehow put some toxic gases into his flat. He brings you some of his vomit to be analyzed, so that his concerns will now be 'demonstrated' and he will be able to take appropriate action against his neighbor .

Questions

- What is your diagnosis ?
- If Mr S start his treatment with Chlorpromazine then his mother noticed ha had developed some repeated involuntary movements, he told you that he had never suffer from it before .
- Write your comment about this and your recommendation .

Answers

- Schizophrenia .
- He had developed Extrapiramidal side effects , Which can be avoided by shifting to Atypical anti psychotic drugs (Risperidone)

Cases

An adolescent male is newly diagnosed with schizophrenia. Which of the following antipsychotic agents may have the best chance to improve his apathy and blunted affect?

- A. Chlorpromazine.
- B. Fluphenazine.
- C. Haloperidol.
- D. Risperidone.
- E. Thioridazine.

Answers

- Correct answer = D. Risperidone is the only antipsychotic on the list that has some reported benefit in improving the negative symptoms of schizophrenia. It is a second-generation antipsychotic, and the other drugs listed are first generation antipsychotic agents. All of the agents have the potential to diminish the hallucinations and delusional thought processes (positive symptoms).

Cases

- Which one of the following antipsychotics has been shown to be a partial agonist at the dopamine D2 receptor?
 - A. Aripiprazole.
 - B. Clozapine.
 - C. Haloperidol.
 - D. Risperidone.
 - E. Thioridazine.

Answers

- Correct answer = A. Aripiprazole is the agent that acts as a partial agonist at D2 receptors. Theoretically, the drug would enhance action at these receptors when there is a low concentration of dopamine and would block the actions of high concentrations of dopamine. All of the other drugs are only antagonistic at D2 receptors, with haloperidol being particularly potent.

Discussions

- **The Dopamine Dilemma: Mechanism vs. Reality**
- **Key Question:** "We know that blocking D2 receptors in the mesolimbic pathway treats positive symptoms, but why does this same action in the nigrostriatal pathway cause EPS? How does this fundamental 'trade-off' shape our entire approach to antipsychotic therapy?"
- **Follow-up/Prompt:** "If you had to explain to a patient why their medication might cause muscle stiffness or restlessness, how would you frame it in simple terms?"

Discussions

- **Case Study Deep Dive: Mr. S (4 minutes)**
- **Key Question:** "Let's revisit Mr. S. Beyond the EPS, what other adverse effects of Chlorpromazine should we be monitoring for, based on its receptor profile (e.g., alpha, muscarinic, histaminic blockade)?"
- **Follow-up/Prompt:** "His mother is now worried about switching to Risperidone after reading online about 'metabolic side effects.' How would you, as the pharmacist, counsel her on balancing the risk of EPS with the risk of weight gain/diabetes?"

Discussions

- **The Pharmacologist's Puzzle:** Partial Agonism vs. Full Antagonism
- **Key Concept:** Moving from the simple "blockade" model to the nuanced "stabilization" model of partial agonism.
- **Key Question:** "We have Aripiprazole and Cariprazine as partial agonists. The lecture says they 'block' high dopamine and 'stimulate' low dopamine. How does this theoretical mechanism translate into practical clinical advantages and disadvantages compared to a pure antagonist like Risperidone?"
- **Follow-up/Prompt:** "Imagine a patient on Risperidone is stable but experiencing significant emotional blunting (a negative symptom). Why might switching to a partial agonist like Aripiprazole be a considered strategy? What new side effect might you warn them about initially?" (This leads them to think about the initial activation or akathisia sometimes seen with partial agonists).

Discussions

- **The Ultimate Clinical Challenge:** Treatment Resistance and Clozapine
- **Key Concept:** Applying knowledge of drug mechanisms and side effect profiles to a high-stakes clinical scenario.
- **Key Question:** "Let's say Mr. S from our case was switched to Risperidone, but after an adequate trial, his positive symptoms (the delusions about his neighbor) persist. The team is now considering Clozapine. Why is Clozapine the gold standard for treatment-resistant schizophrenia, and what monumental responsibility does this place on the pharmacist?"
- **Follow-up/Prompt:** "The slide mentions 'leukopenia: often fatal.' What specific, concrete monitoring protocol must the pharmacist ensure is in place before and during Clozapine therapy? What would you tell the patient and their family is the most important reason for these weekly blood tests?"